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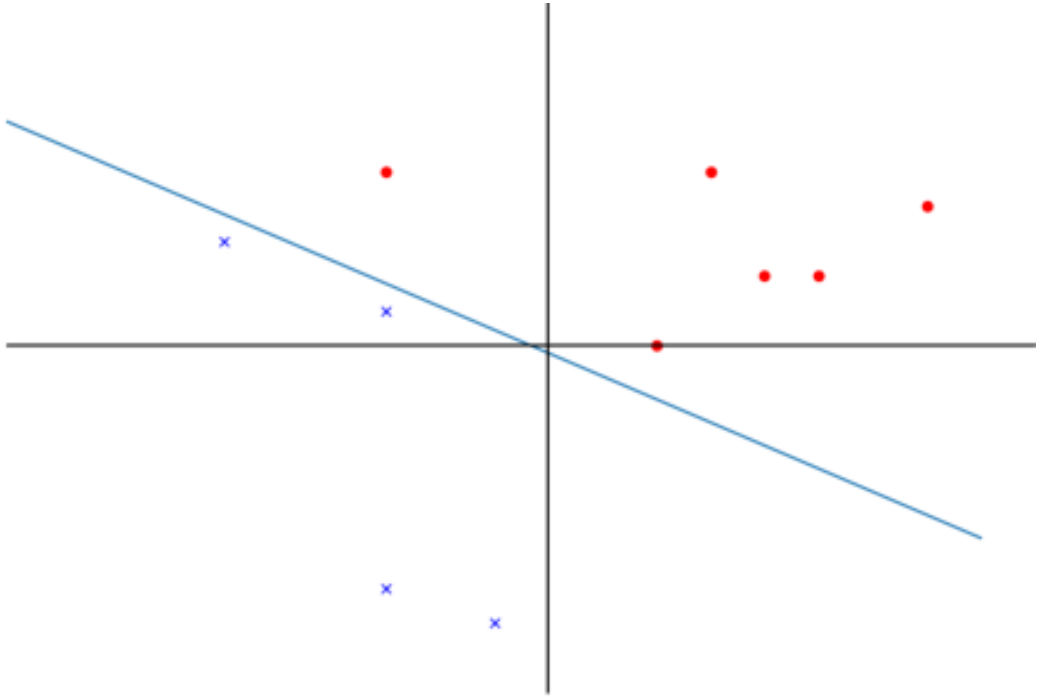
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Lecture VII – MultiLayer Perceptrons

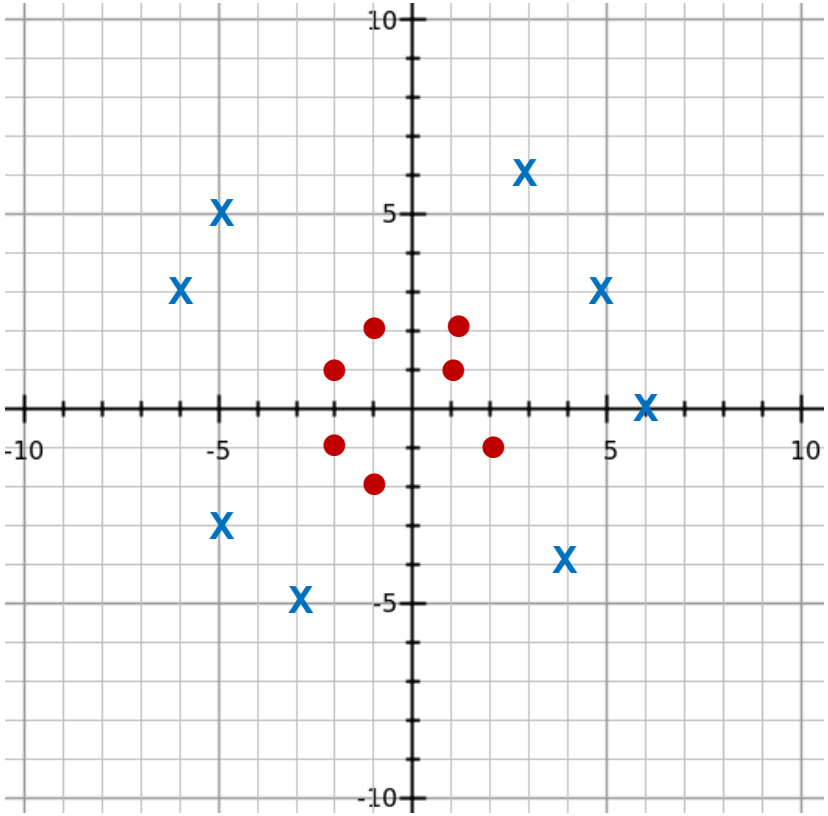
Machine Learning for Business

Michael Deamer

Perceptron Example

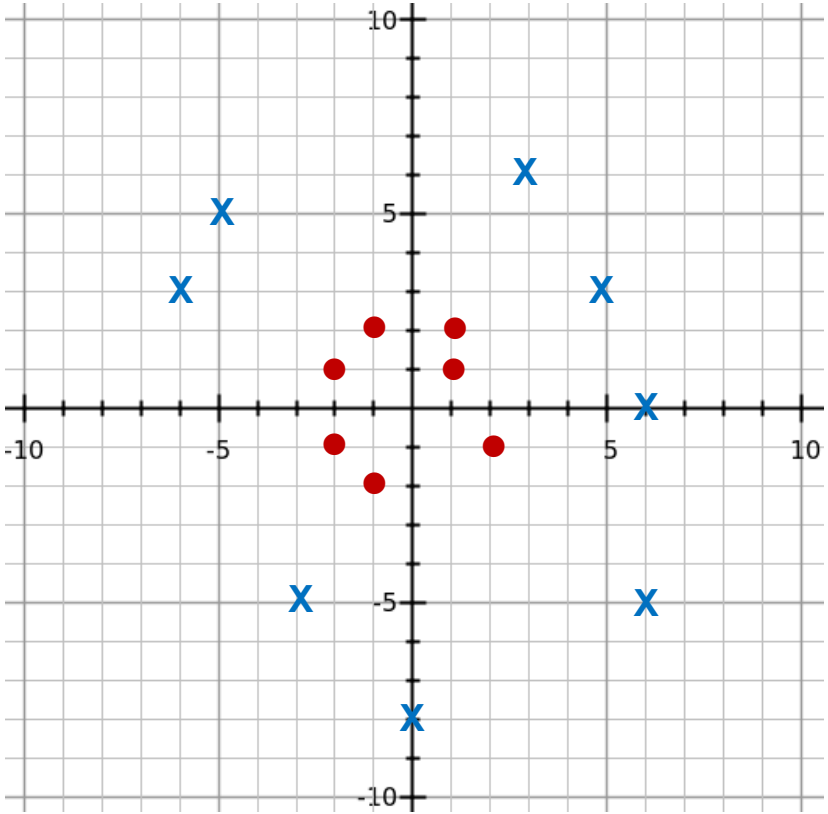


Example Data



Example Data

X	Y	Color
6	0	Blue
6	-5	Blue
5	3	Blue
3	6	Blue
2	-1	Red
1	1	Red
1	2	Red
-1	2	Red
-1	-2	Red
-2	1	Red
-2	-1	Red
-3	-5	Blue
-5	5	Blue
-5	-8	Blue
-6	3	Blue



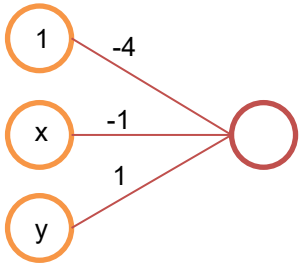
Example Data

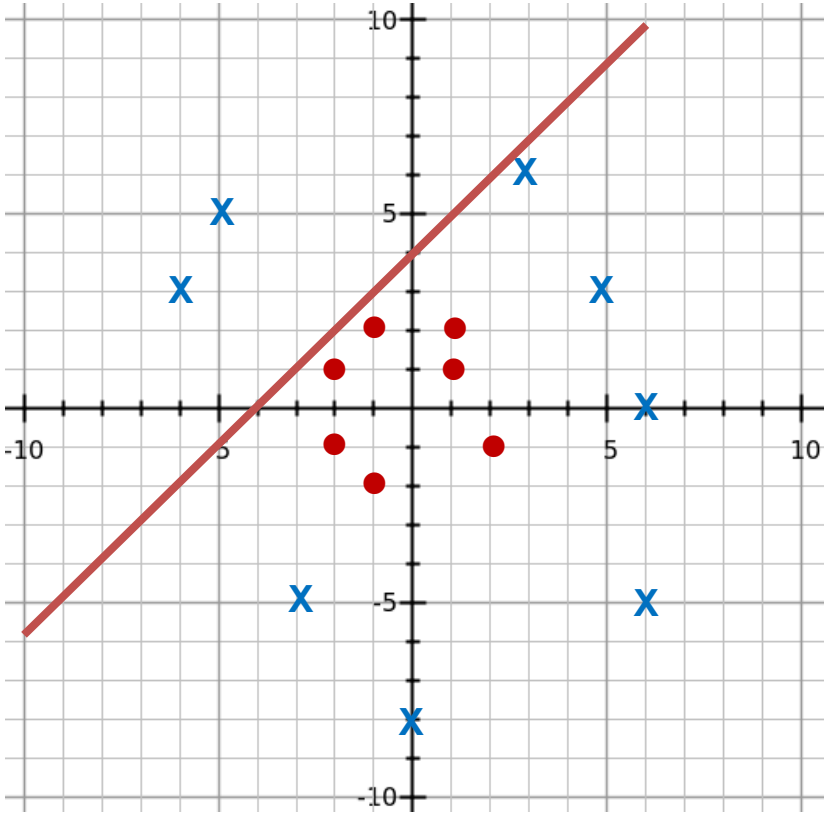
$$\omega = \begin{bmatrix} -4 \\ -1 \\ 1 \end{bmatrix}$$

ωTx
-4 + -1x + 1y

Line Equation
y = 1x + 4

Perceptron





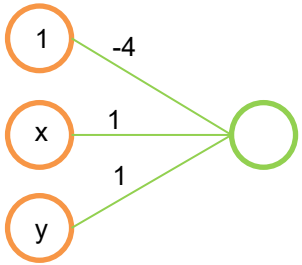
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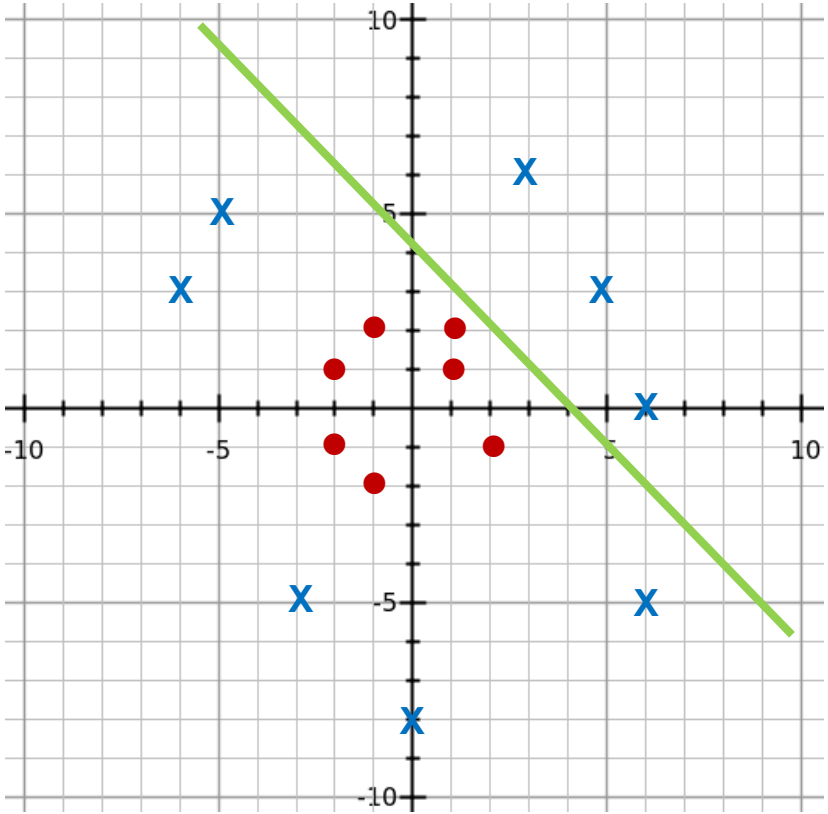
$$\omega = \begin{bmatrix} -4 \\ 1 \\ 1 \end{bmatrix}$$

ωTx
-4 + 1x + 1y

Line Equation
y = -1x + 4

Perceptron





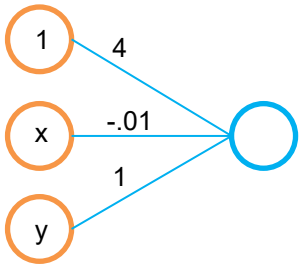
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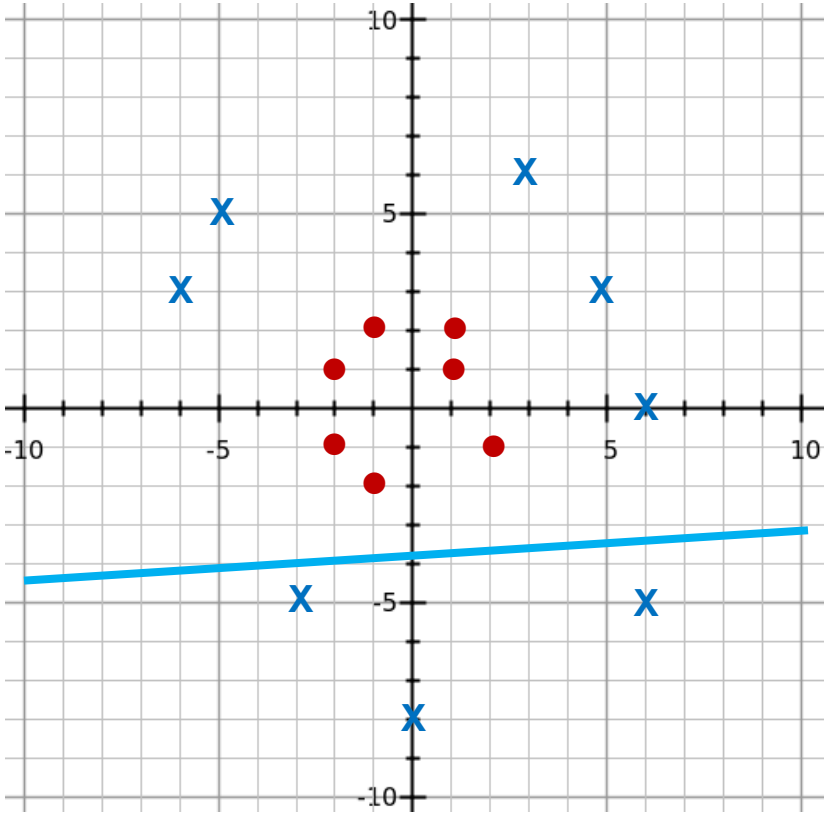
$$\omega = \begin{bmatrix} 4 \\ -.01 \\ 1 \end{bmatrix}$$

ωTx
4 + -.01x + 1y

Line Equation
y = .01x - 4

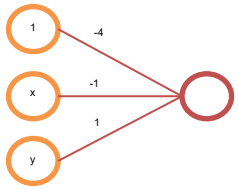
Perceptron



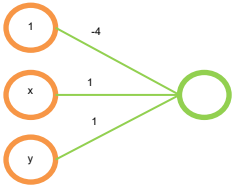


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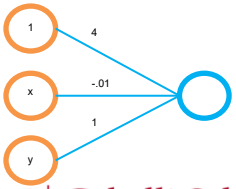
Perceptron 1

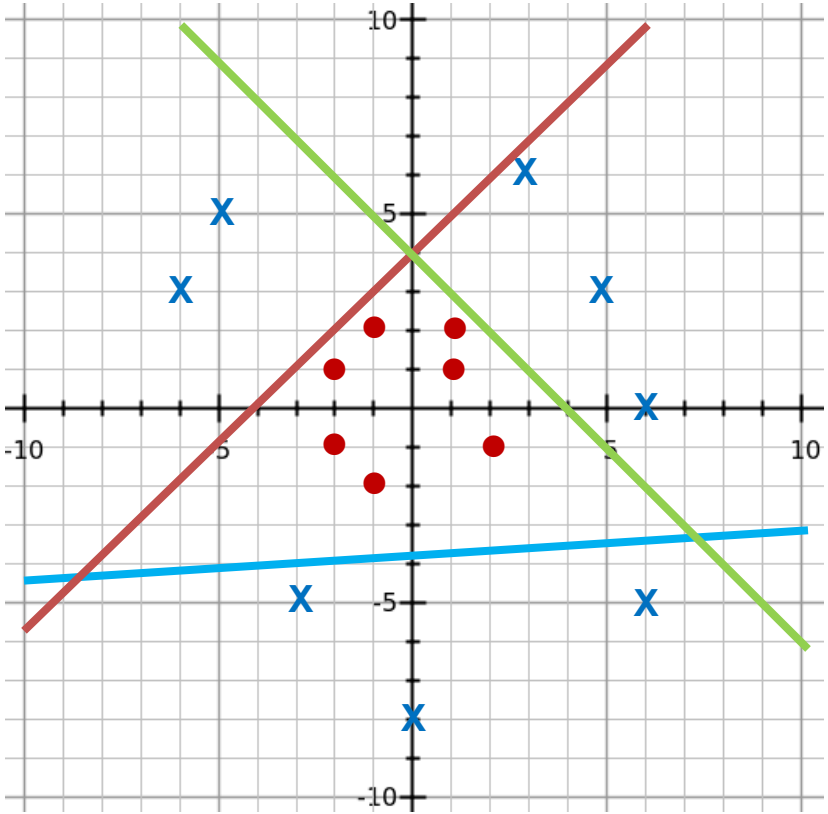


Perceptron 2



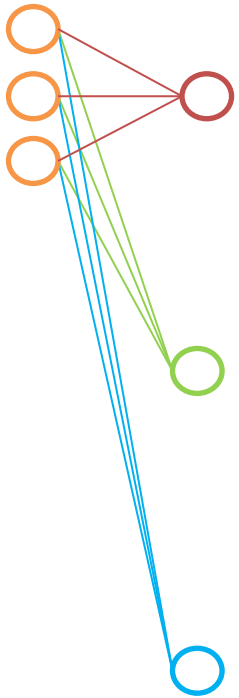
Perceptron 3

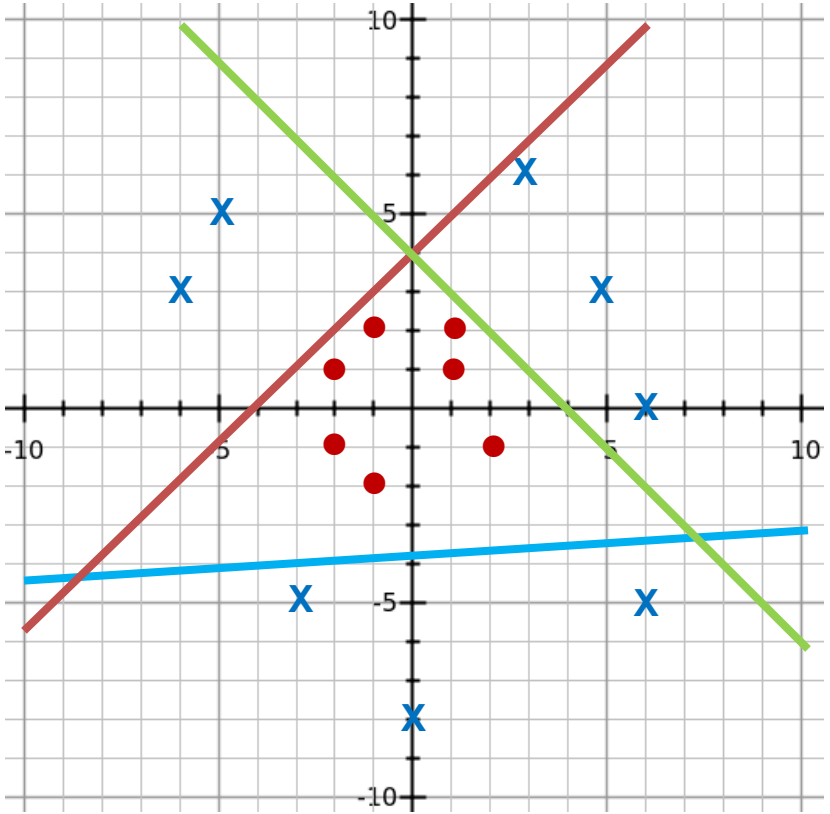




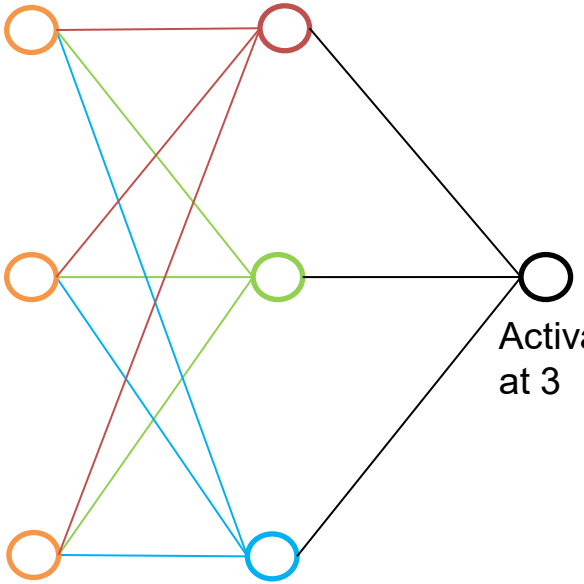
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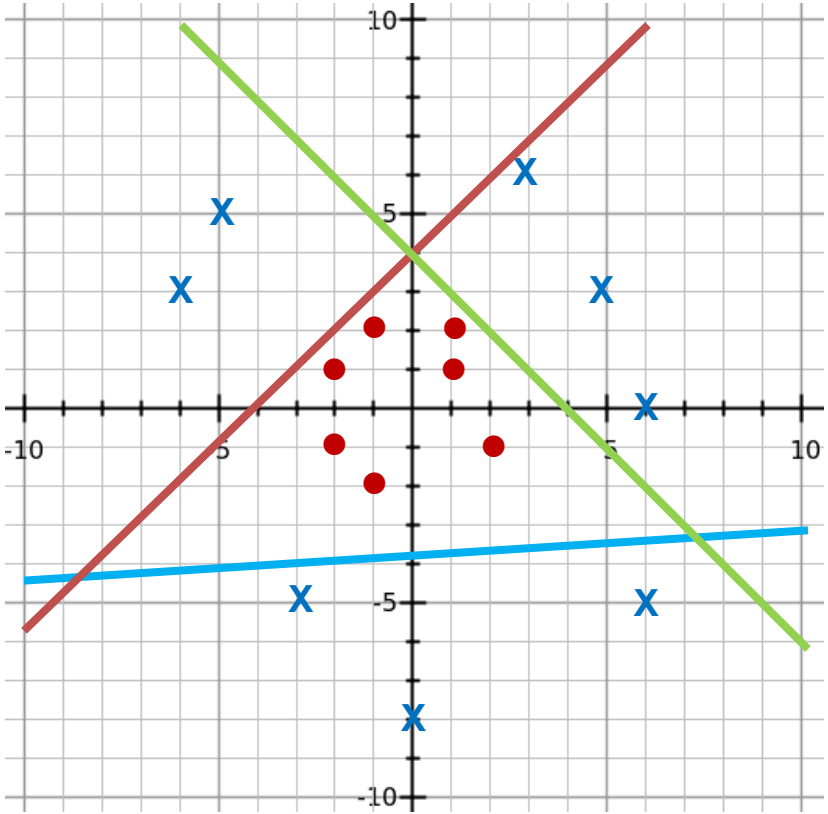
Example Data



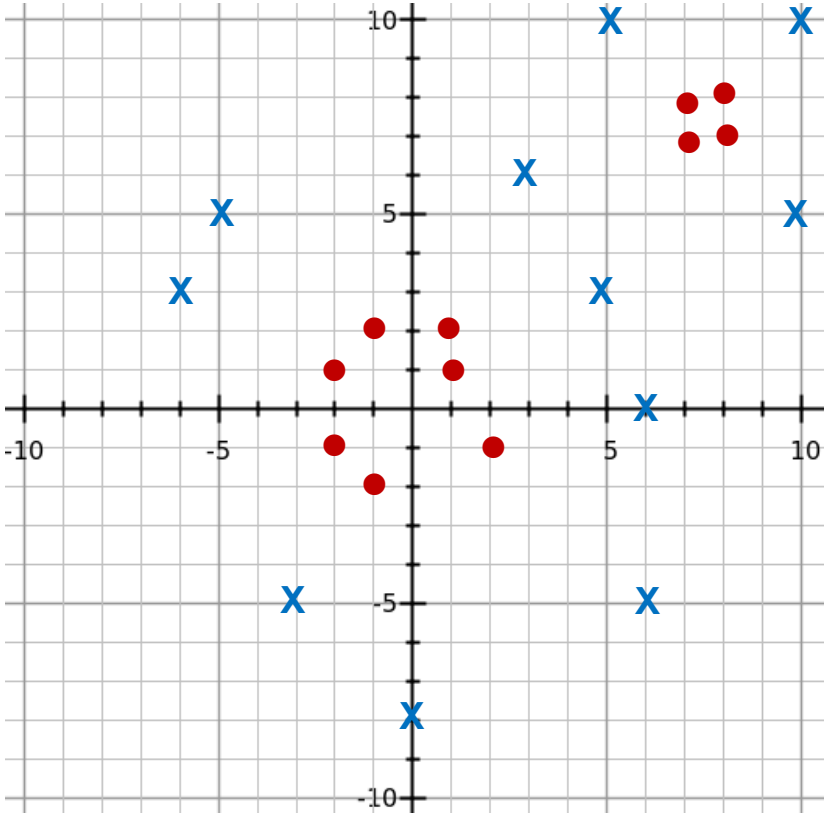


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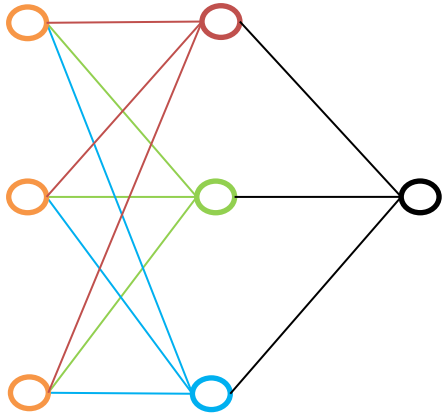


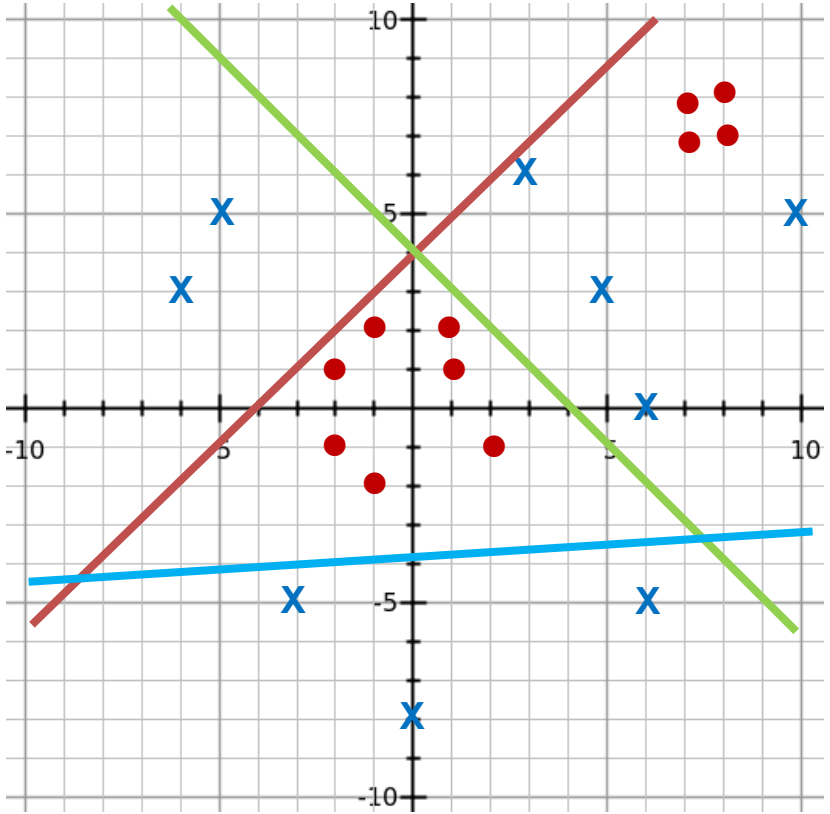


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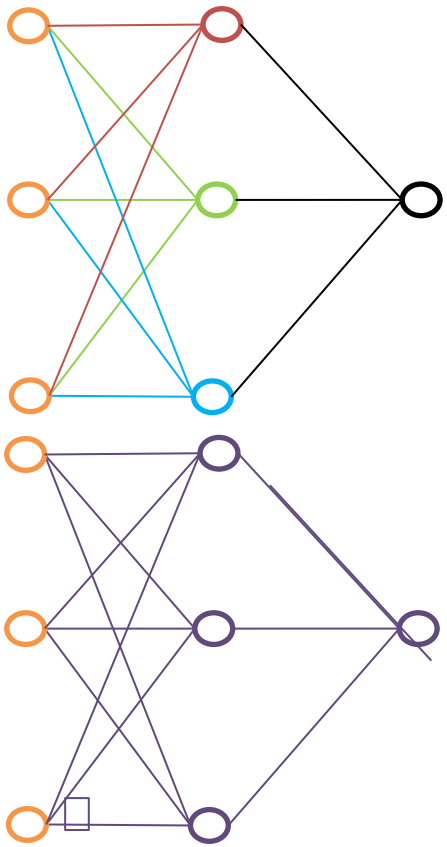


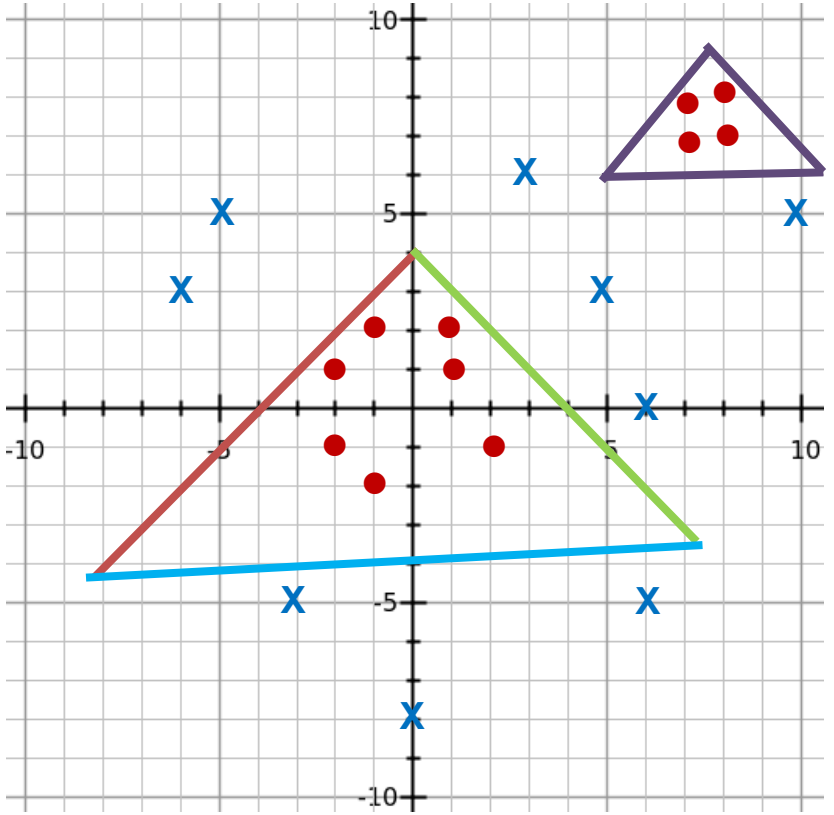
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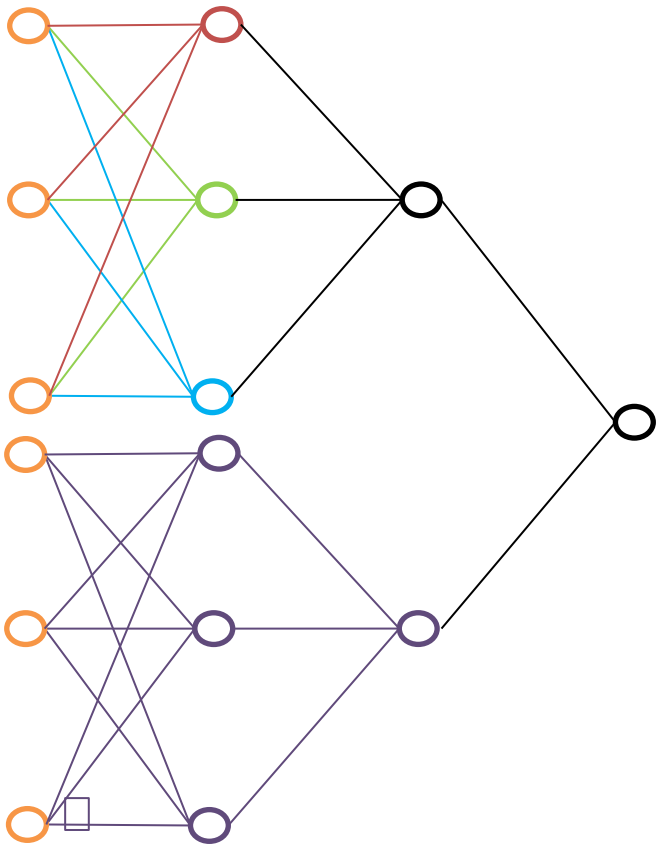


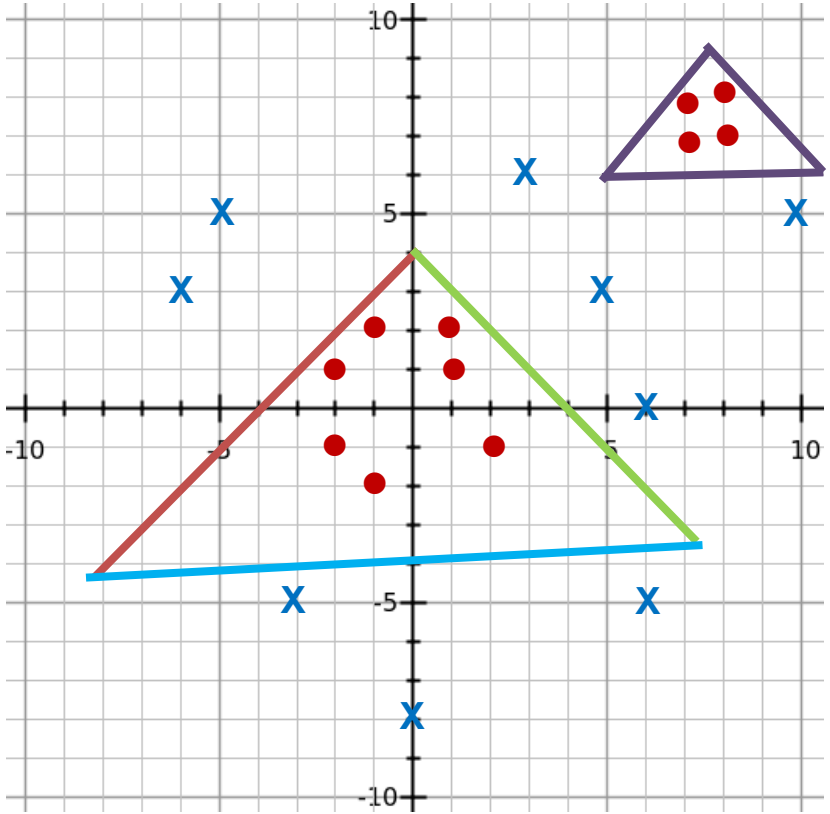
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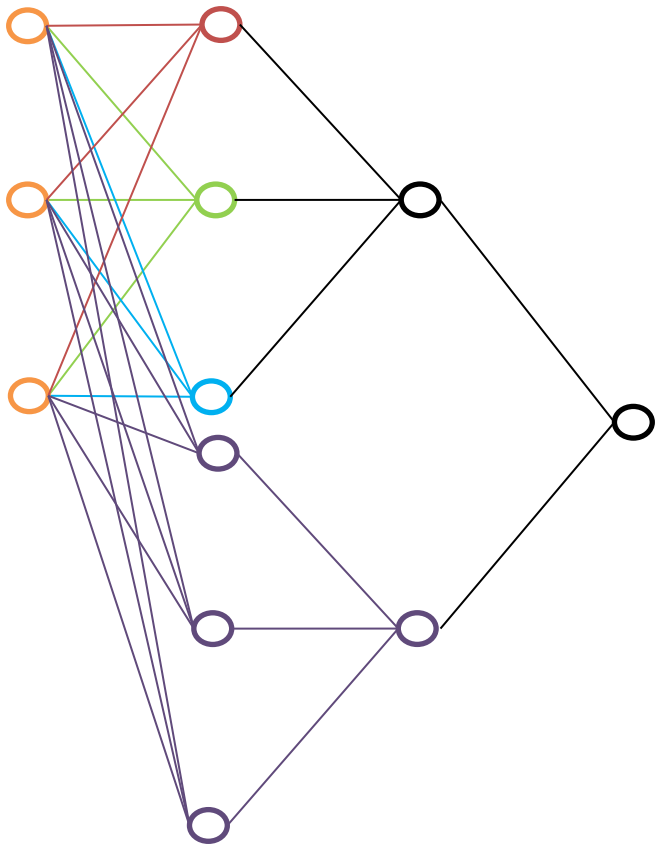


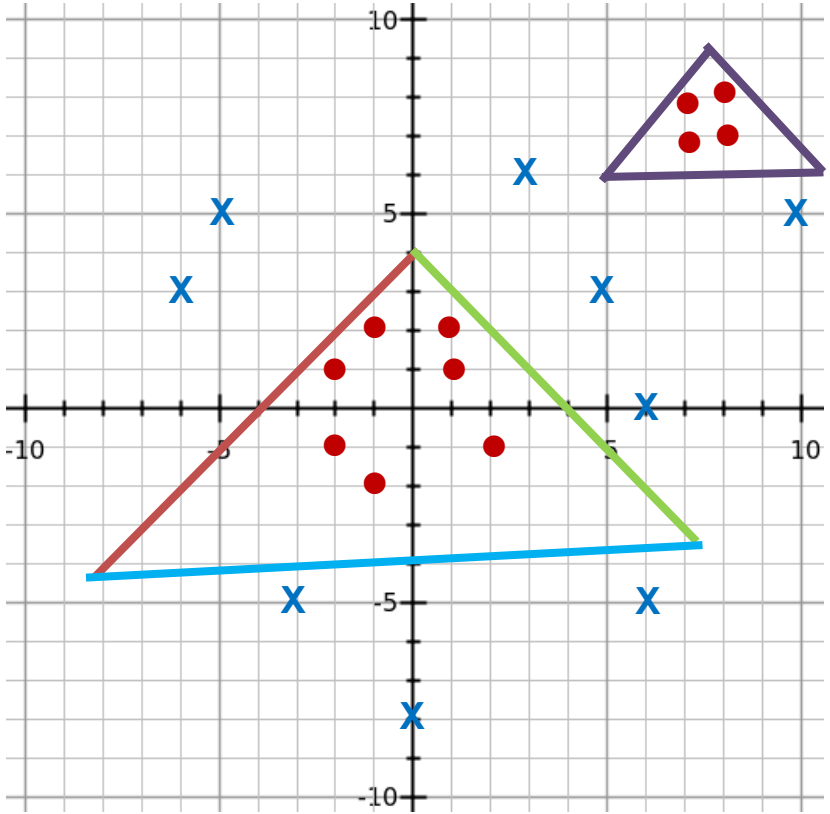
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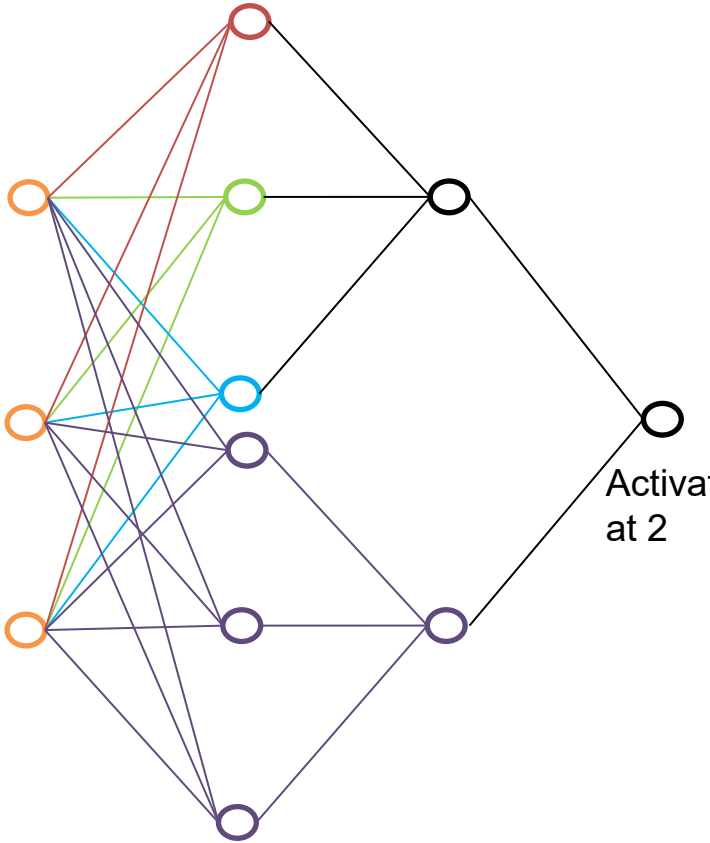


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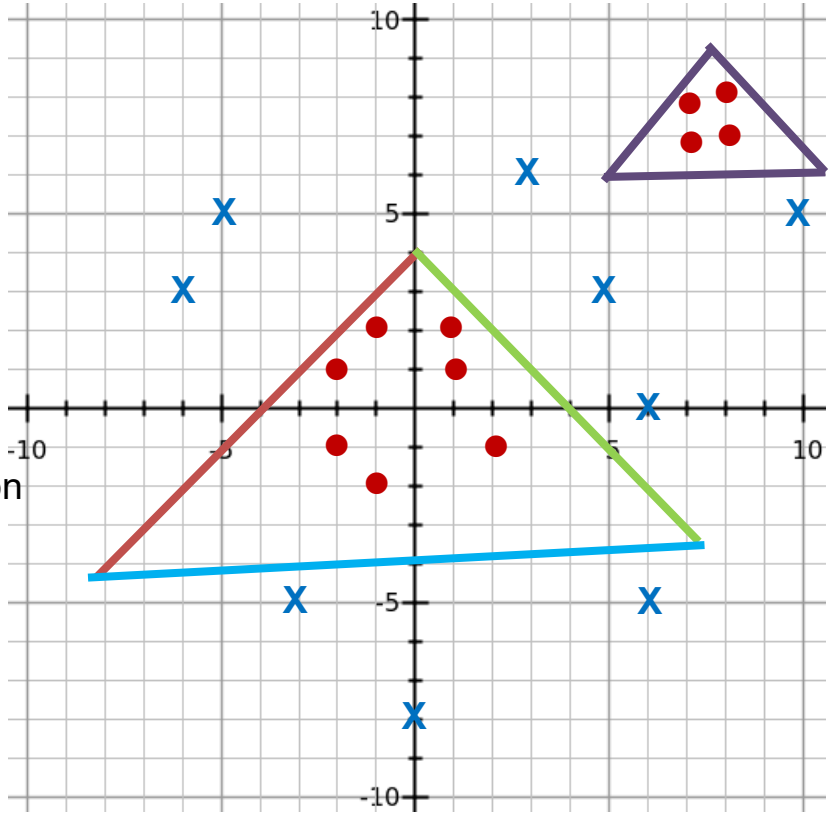




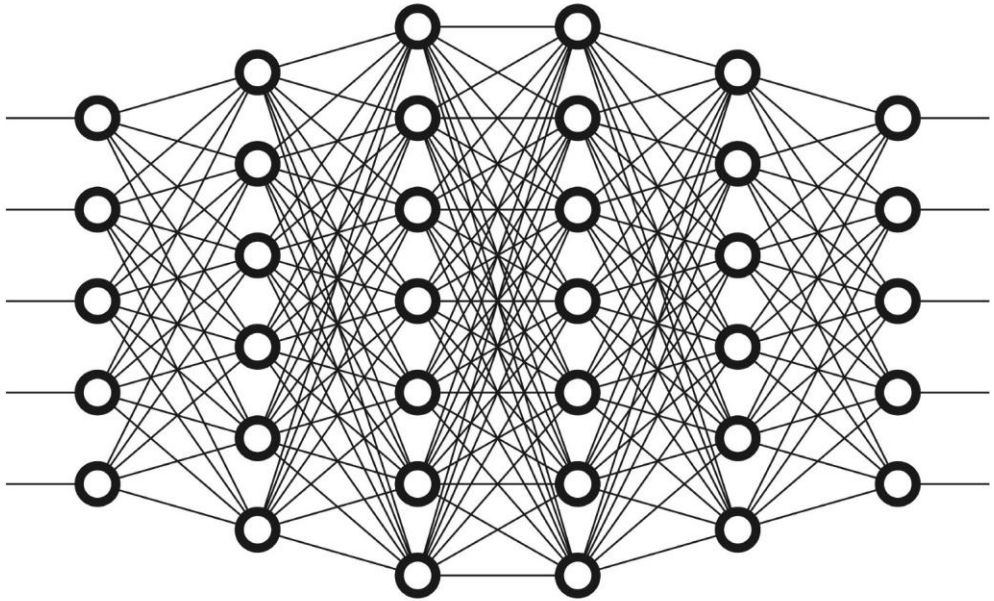
Example Data



Activation at 2



Hidden Layers



Edges



Edges





Edges







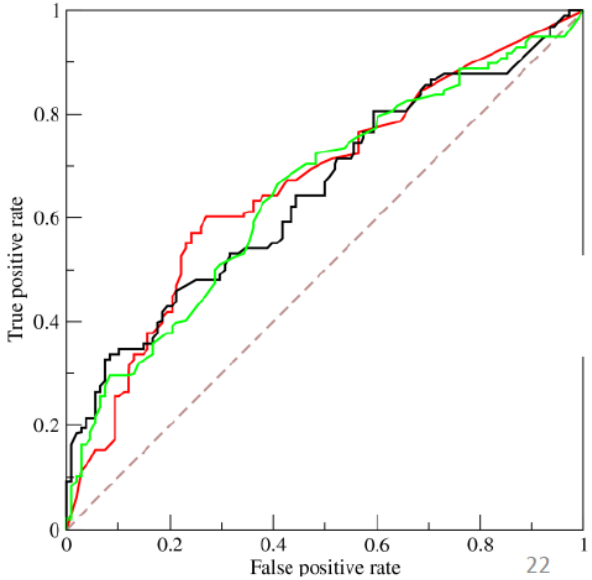
Edges



ROC and AUC

- Commonly used to present results for binary decision problem in Machine Learning.
- Receiver operating characteristic (ROC) is a graphical plot of true positive rate against false positive rate.
- Cost/benefit analysis of decision.
- Relative tradeoffs between TP

(benefit) and FP (costs) ratio.



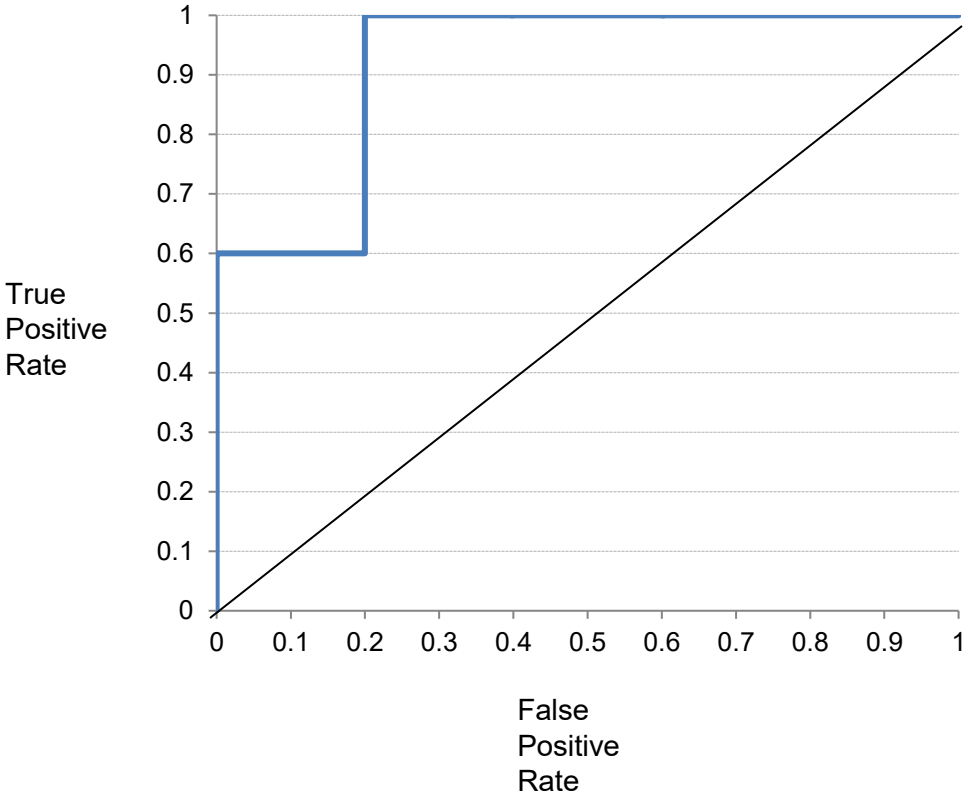
ROC Example

Person	Actual Eligible	Probability of Eligible	Prediction
David	Eligible	91	Eligible
Patrick	Eligible	81	Eligible
Larry	Eligible	71	Eligible
Scottie	Not Eligible	61	Eligible
Michael	Eligible	51	Eligible
Clyde	Eligible	41	Eligible
Karl	Not Eligible	31	Eligible
Magic	Not Eligible	21	Eligible
Charles	Not Eligible	11	Eligible
John	Not Eligible	1	Eligible

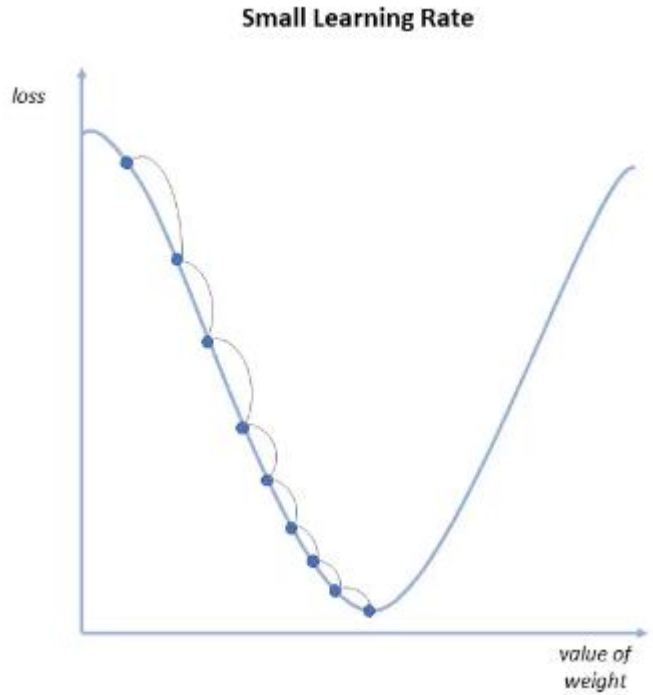
ROC Example

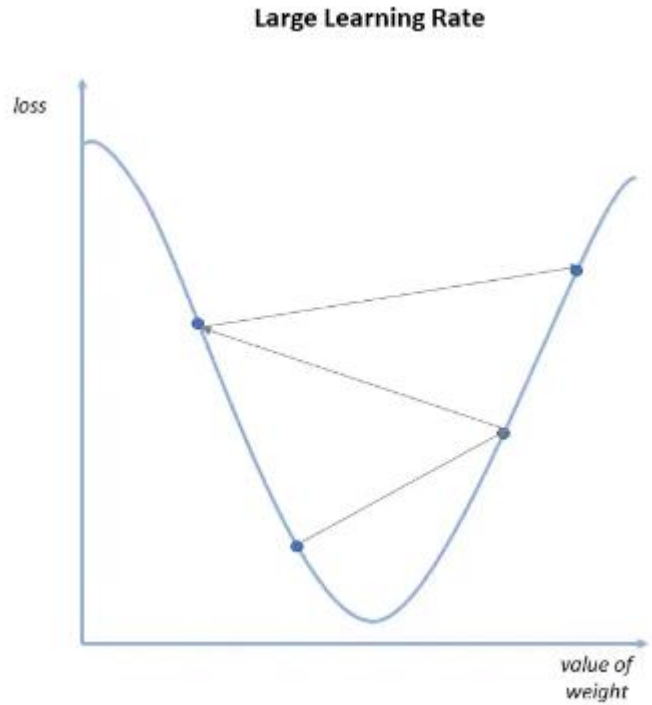
Thresholds

	TP	FP
0	1	1
10	1	0.8
20	1	0.6
30	1	0.4
40	1	0.2
50	0.8	0.2
60	0.6	0.2
70	0.6	0
80	0.4	0
90	0.2	0
100	0	0

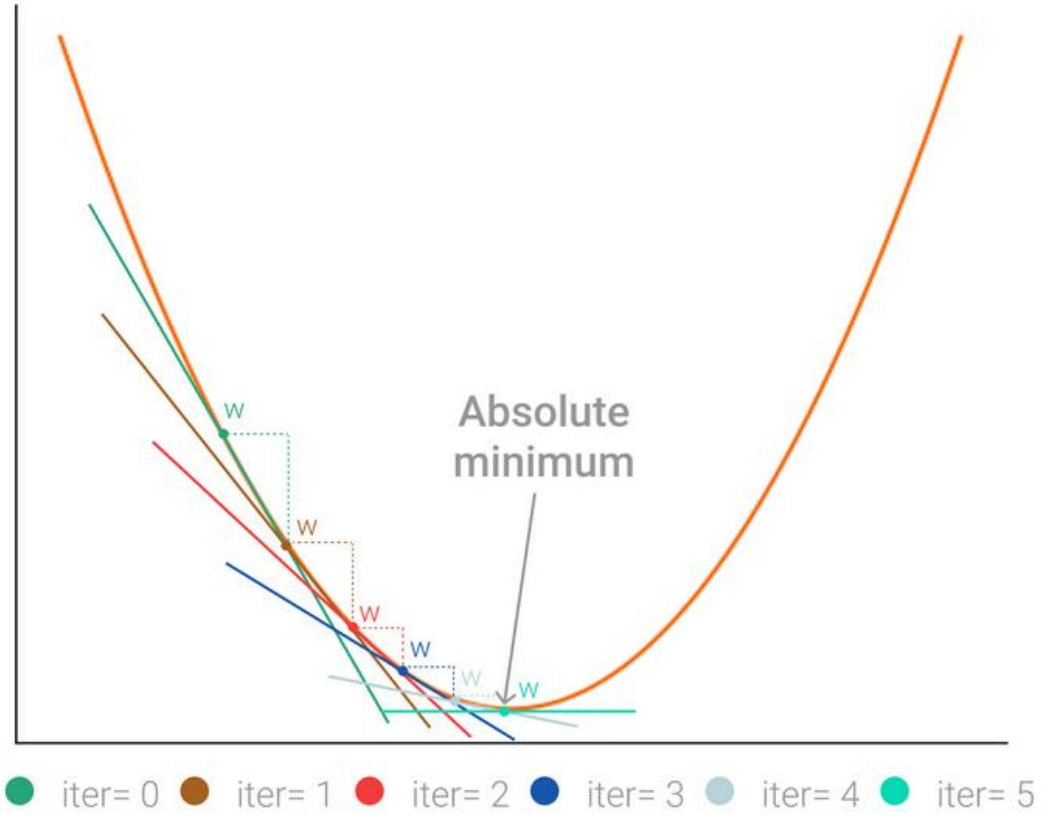


Gradient Descent

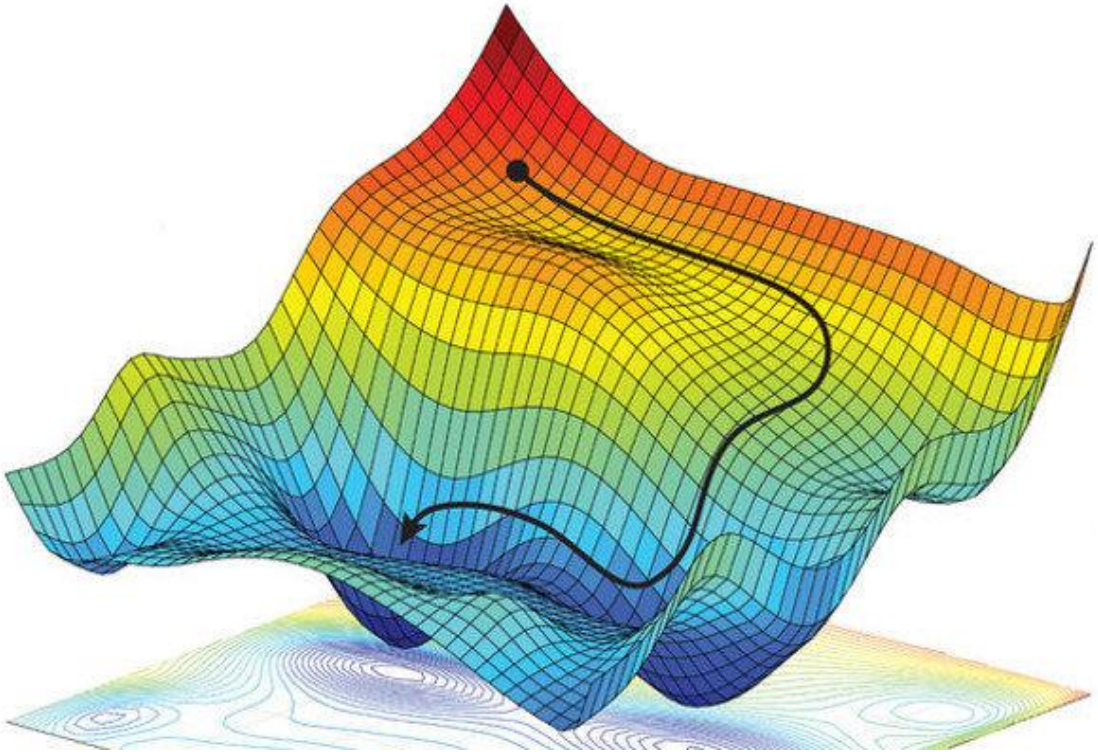




Gradient Descent

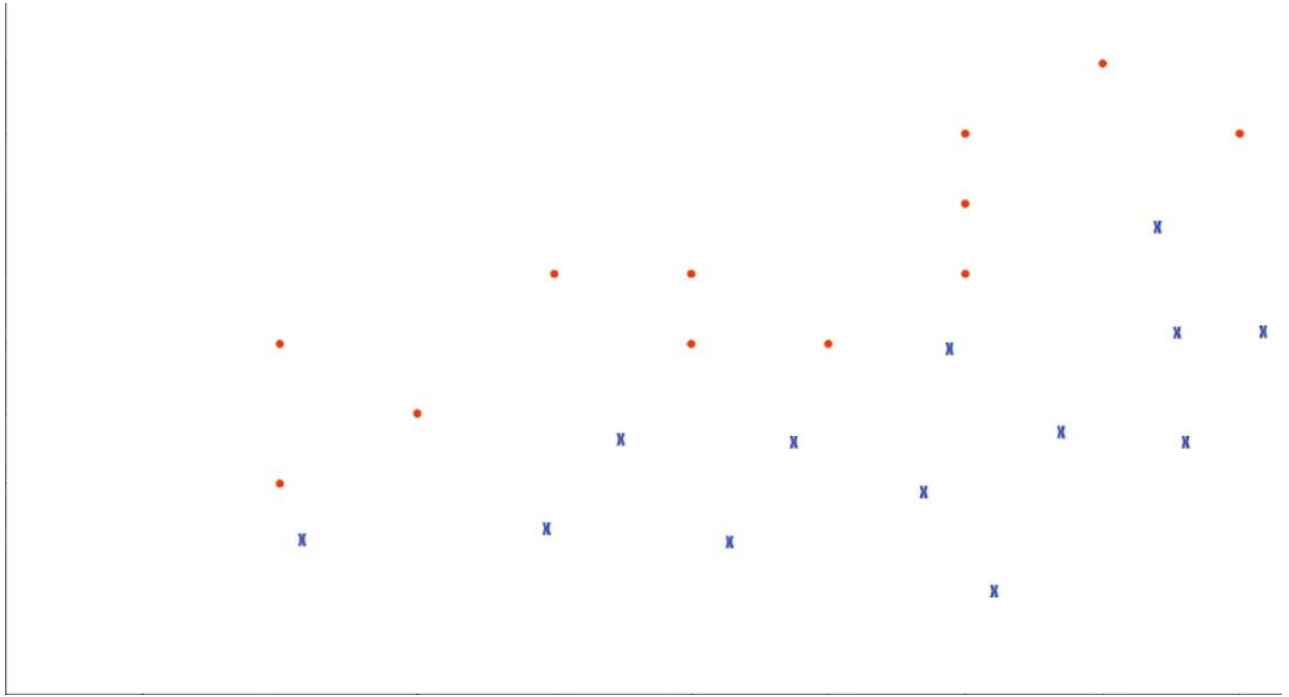


Gradient Descent



Limitation of Decision tree

Interaction between variables



To Do

- Homework VI and Lab VI
- Recommended reading: ROC Curve 6.6.6; Bayes Nets 6.3 and 7.2
- Gradient Descent explanation by IBM

Bibliography

Tan, Steinbach, Karpatne, and Kumar. Introduction to Data Mining, second edition. Pearson, Hudson, NY, 2019.
Han, Kamber, Pei. Data Mining Concepts and Techniques, second edition. University of Illinois at Urbana-Champaign. Elsevier, San Francisco, CA, 2006.
Bento, Carolina. Stochastic Gradient Descent explained in real life